

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

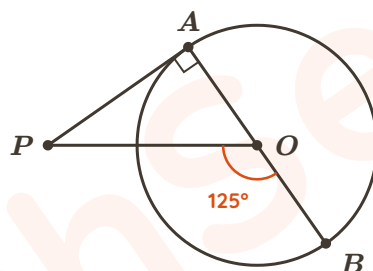
Chapter 10: Circles

Previous year board questions, topic-wise: 27 questions

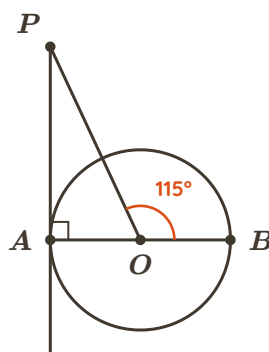
The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Theorem 1: tangent and radius

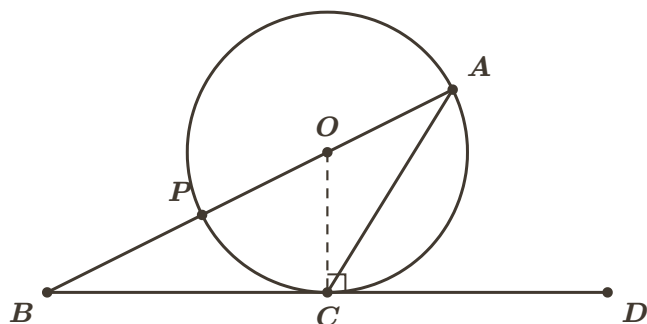
1. PA is a tangent from an external point P to a circle with centre O , touching it at A . AOB is a diameter and $\angle POB = 125^\circ$. Find $\angle APO$. (a) 25° (b) 65° (c) 90° (d) 35° CBSE 2026, 1 mark



2. Two concentric circles have radii 5 cm and 4 cm. A chord of the larger circle touches the smaller circle. Find the length of the chord. CBSE 2026, 2 marks
3. Tangents are drawn to a circle at the two ends of a diameter. How are the two tangents related? CBSE 2025, 1 mark
4. AOB is a diameter of a circle with centre O , and PA is the tangent at A . If $\angle POB = 115^\circ$, find $\angle APO$. CBSE 2025, 1 mark



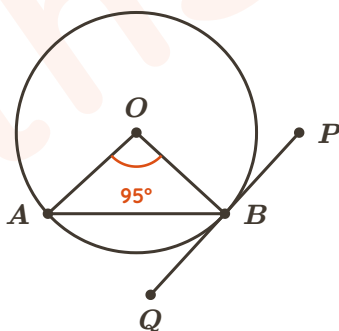
5. O is the centre of a circle and BCD is the tangent to it at C . The line from B through O meets the circle at P and A . Prove that $\angle BAC + \angle ACD = 90^\circ$. **CBSE 2025, 2 marks**



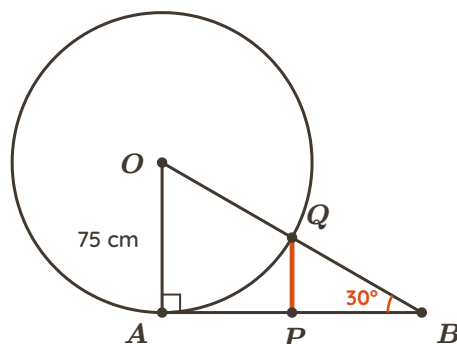
6. Assertion (A): the tangents at the two ends of a diameter of a circle are parallel. Reason (R): the diameter is the longest chord of a circle. Choose: (a) A and R are both true and R explains A. (b) A and R are both true but R does not explain A. (c) A is true and R is false. (d) A is false and R is true.

CBSE 2024, 1 mark

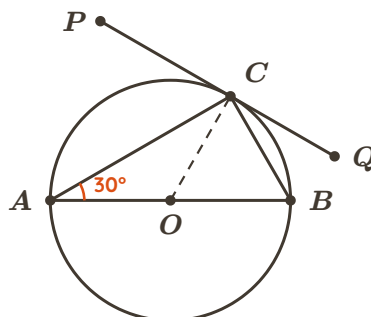
7. PQ is the tangent at B to a circle with centre O , and AB is a chord with $\angle AOB = 95^\circ$. Q lies on the same side of OB as A . Find $\angle ABQ$. (a) 47.5° (b) 42.5° (c) 85° (d) 95° **CBSE 2023, 1 mark**



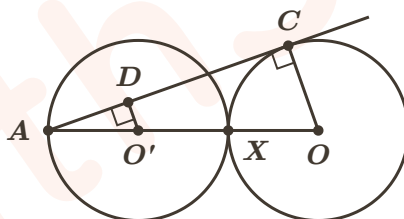
8. A discus leaves the thrower's circular path along a tangent. AB is such a tangent, touching a circle of radius 75 cm and centre O at A , with $\angle ABO = 30^\circ$. OB cuts the circle at Q , and P is the point on AB with $PQ \parallel OA$. (i) Find AB . (ii) Find OB . (iii) Find AP , OR find PQ . **CBSE 2023, 4 marks**



9. Two concentric circles have radii 4 cm and 3 cm. Find the length of the chord of the larger circle that touches the smaller circle. **CBSE 2022 Term 2, 2 marks**
10. PQ is the tangent at C to a circle with centre O , and AB is a diameter with $\angle CAB = 30^\circ$. P lies on the same side of OC as A . Find $\angle PCA$. **CBSE 2016, 1 mark**

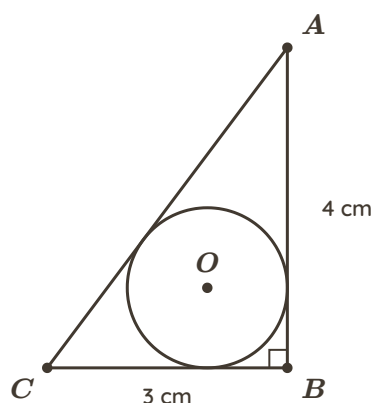


11. Two equal circles with centres O and O' touch each other at X . The line OO' produced meets the circle with centre O' again at A . AC is a tangent from A to the circle with centre O , touching it at C , and $O'D \perp AC$. Find $\frac{DO'}{CO}$. **CBSE 2016, 4 marks**

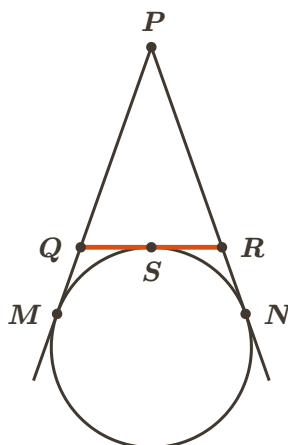


2 Theorem 2: tangents from an external point

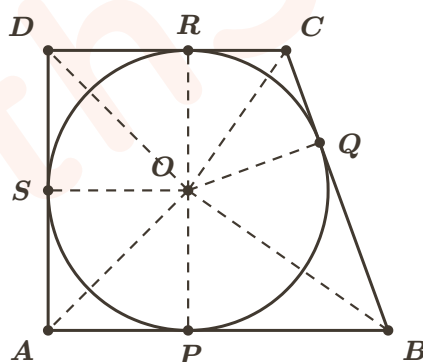
12. TP and TQ are tangents from an external point T to a circle with centre O , and $\angle POQ = 120^\circ$. Find $\angle PTQ$. (a) 60° (b) 70° (c) 80° (d) 90° **CBSE 2026, 1 mark**
13. $\triangle ABC$ is right-angled at B with $AB = 4$ cm and $BC = 3$ cm. Find the radius of the circle inscribed in the triangle. **CBSE 2026, 3 marks**



14. A circle touches the side QR of $\triangle PQR$ at S , and touches PQ and PR produced at M and N .
Prove that $PM = \frac{1}{2}(PQ + QR + PR)$. **CBSE 2026, 3 marks**

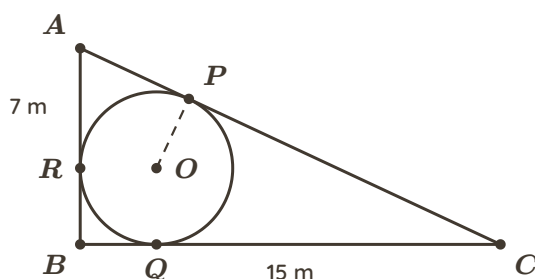


15. A point P is 26 m from the centre of a circle. Each tangent from P to the circle is 10 m long. Find the radius of the circle. **CBSE 2025, 2 marks**
16. A quadrilateral is drawn around a circle so that all four sides touch it. Prove that each pair of opposite sides subtends supplementary angles at the centre. **CBSE 2025, 2 marks**

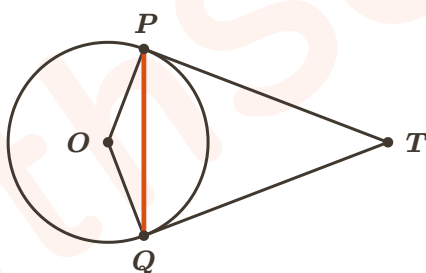


17. AB is a diameter of a circle with centre O . The tangents at A and B are met by a third tangent at Q and P respectively. Prove that $\angle POQ = 90^\circ$. **CBSE 2024, 3 marks**
18. A circle of radius 8 cm with centre O touches the sides AD , DC , CB and BA of a quadrilateral $ABCD$ at P , Q , R and S . $AD \perp DC$, $BC = 30$ cm and $BS = 24$ cm. Find DC . **CBSE 2024, 3 marks**

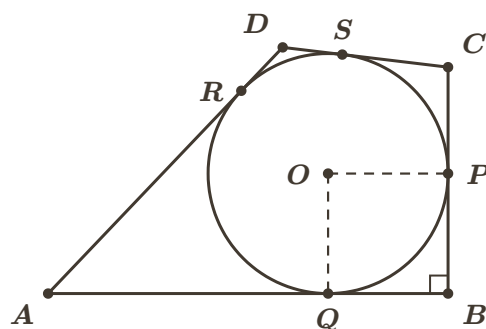
19. A triangular backyard ABC has a right angle at B , with $AB = 7$ m and $BC = 15$ m. A circular pit inside it touches AC , BC and AB at P , Q and R , and $AP = x$ m. (i) Write AR in terms of x . (ii) What kind of quadrilateral is $BQOR$? (iii) Write PC in terms of x and find x , OR find x and then the radius r of the pit. **CBSE 2024, 4 marks, case study**



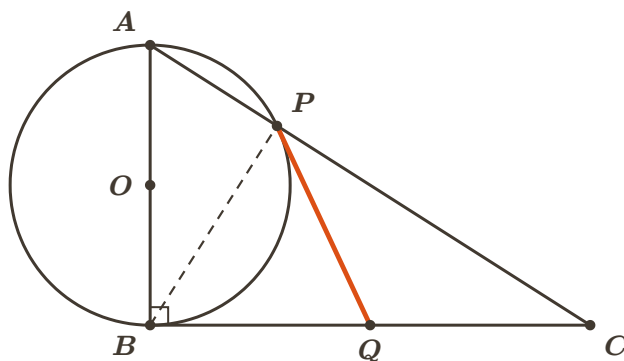
20. TP and TQ are tangents from an external point T to a circle with centre O . Prove that $\angle PTQ = 2\angle OPQ$. **CBSE 2023, 3 marks**



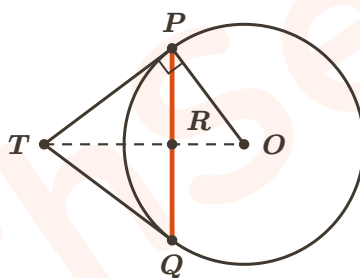
21. A circle with centre O is inscribed in a quadrilateral $ABCD$, touching AB , BC , CD and DA at Q , P , S and R . $\angle B = 90^\circ$, $AD = 17$ cm, $AB = 20$ cm and $DS = 3$ cm. Find the radius of the circle. **CBSE 2023, 3 marks**



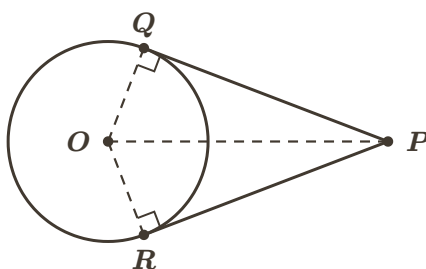
22. $\triangle ABC$ is right-angled at B . A circle drawn on AB as diameter cuts AC at P . The tangent to the circle at P meets BC at Q . Prove that Q is the mid-point of BC . CBSE 2022 Term 2, 4 marks



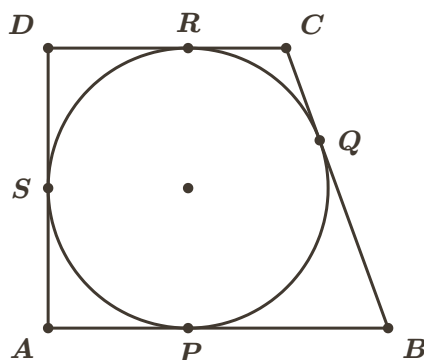
23. PQ is a chord of length 8 cm in a circle of radius 5 cm with centre O . The tangents at P and Q meet at T . Find TP . CBSE 2019, 3 marks



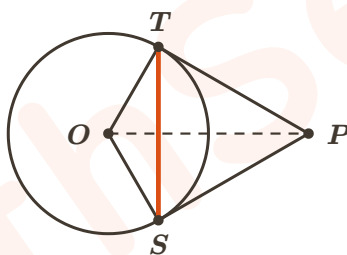
24. Prove that the two tangents drawn to a circle from a point outside it are equal in length. CBSE 2018, 3 marks



25. A quadrilateral $ABCD$ is drawn around a circle, with AB , BC , CD and DA touching the circle at P , Q , R and S . Prove that $AB + CD = BC + DA$. **CBSE 2016, 2 marks**



26. PT and PS are tangents from an external point P to a circle with centre O and radius r . If $OP = 2r$, show that $\angle OTS = \angle OST = 30^\circ$. **CBSE 2016, 2 marks**



27. Prove that the tangents drawn to a circle from an external point are equal in length. **CBSE 2016, 4 marks**

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